A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-18397407})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary ideal I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{\,(1-\sigma_x)^2}\,(t)\,J\mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 1884 & 1639 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,1884 + \mathbf{Z}\,(1639 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 4599352 = 0$, of discriminant -18397407. Class group invariants $(40 \quad 10 \quad 5)$, class number 2000; the three stored generator ideals (HNF) are $\begin{pmatrix} 62 & 7 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1039 & 479 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 153y^3 + (5s - 49061)\, y^2 + (5s + 339709)\, y + (-9460s - 16257223)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 310*y^8 - 98729*y^7 + 899066*y^6 - 48894653*y^5 + 269
0719747*y^4 - 38077663276*y^3 + 1275994253977*y^2 - 11483843568669*y
+ 676054462444509
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad 1 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right) s, \qquad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1364932699083 = 3^3 \cdot 6101 \cdot 8286029$:

$$\begin{pmatrix}
151659188787 & 92029726837 & 9034141479 & 16054093719 & 81654813653 & 53887090936 & 74422383738 & 55978661180 & 27157940536 & 53924866653 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 3 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 3 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 380 factors with signed exponents of absolute value up to 12563867541167585379; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 3 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 153y^3 + (5s - 49061)\,y^2 + (5s + 339709)\,y + (-9460s - 16257223)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 310*y^8 - 98729*y^7 + 899066*y^6 - 48894653*y^5 + 269
0719747*y^4 - 38077663276*y^3 + 1275994253977*y^2 - 11483843568669*y
+ 676054462444509
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad 1 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right) s, \qquad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4136948743179 = 3 \cdot 7^3 \cdot 4020358351$:

$$\begin{pmatrix}
84427525371 & 32662630560 & 82642821275 & 50040169739 & 12392051524 & 68065394625 & 74336205091 & 8005853897 & 43250118945 & 72115595837 \\
0 & 7 & 0 & 1 & 5 & 2 & 1 & 0 & 1 & 3 \\
0 & 0 & 7 & 6 & 1 & 1 & 6 & 3 & 3 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 153y^3 + (5s - 49061)\, y^2 + (5s + 339709)\, y + (-9460s - 16257223)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 310*y^8 - 98729*y^7 + 899066*y^6 - 48894653*y^5 + 269
0719747*y^4 - 38077663276*y^3 + 1275994253977*y^2 - 11483843568669*y
+ 676054462444509
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad 1 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + (-\frac{59454}{16272883861333507})\, s, \qquad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1364932699083 = 3^3 \cdot 6101 \cdot 8286029$:

$$\begin{pmatrix}
151659188787 & 92029726837 & 9034141479 & 16054093719 & 81654813653 & 53887090936 & 74422383738 & 55978661180 & 27157940536 & 53924866653 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 3 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 3 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 385 factors with signed exponents of absolute value up to 5407207007136419622; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 3 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 381y^3 + (-9s - 52315)\, y^2 + (-357s - 587577)\, y + (-2838s - 10530411)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 762*y^8 - 104639*y^7 - 1030350*y^6 + 18803799*y^5 + 355774726
3*y^4 + 99082836570*y^3 + 2268632753133*y^2 + 21700095645411*y + 147
963744385227
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + (-\frac{7859}{2637314866215936})\, s, \qquad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1445520196512 = 2^5 \cdot 3^6 \cdot 7 \cdot 8852147$:

$$\begin{pmatrix}
3346111566 & 2214016038 & 1581849742 & 2259694972 & 3195991622 & 1600146428 & 536938132 & 2945984644 & 3191610299 & 2703188219 \\
0 & 54 & 12 & 42 & 50 & 23 & 45 & 31 & 34 & 39 \\
0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 383 factors with signed exponents of absolute value up to 529906430669; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 381y^3 + (-9s - 52315)\,y^2 + (-357s - 587577)\,y + (-2838s - 10530411)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 762*y^8 - 104639*y^7 - 1030350*y^6 + 18803799*y^5 + 355774726
3*y^4 + 99082836570*y^3 + 2268632753133*y^2 + 21700095645411*y + 147
963744385227
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + (-\frac{1443}{474207141581824})\, s, \qquad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $205522872989 = 7 \cdot 67 \cdot 438215081$:

$$\begin{pmatrix}
205522872989 & 99708957569 & 82125197537 & 178481675936 & 76460983232 & 22964376023 & 153333158652 & 198282447692 & 119271531358 & 65204618837 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 384 factors with signed exponents of absolute value up to 586913612429; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 381y^3 + (-9s - 52315)\, y^2 + (-357s - 587577)\, y + (-2838s - 10530411)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 762*y^8 - 104639*y^7 - 1030350*y^6 + 18803799*y^5 + 355774726
3*y^4 + 99082836570*y^3 + 2268632753133*y^2 + 21700095645411*y + 147
963744385227
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + (-\frac{59454}{16272883861333507})\, s, \qquad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2658403028583 = 3^3 \cdot 13597 \cdot 7241257$:

$$\begin{pmatrix}
886134342861 & 519340949357 & 778399074402 & 728222445557 & 449587310514 & 332301010764 & 549781219118 & 857932609108 & 622628702979 & 244313521912 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 377 factors with signed exponents of absolute value up to 200806330069; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s - 482)\, y^3 + (20s - 44079)\, y^2 + (35s + 713966)\, y + (-1386s - 1930467)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 949*y^8 - 84278*y^7 + 6612677*y^6 - 151021673*y^5 + 2
786318041*y^4 - 41877012822*y^3 + 430627163226*y^2 - 3203861858505*y
+ 12564715259943
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + \left(-\frac{7859}{2637314866215936}\right) s, \qquad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2081607454119 = 3 \cdot 693869151373$:

$$\begin{pmatrix} 2081607454119 & 1835215672271 & 996348053807 & 1346122103099 & 1120149904008 & 1632683719257 & 1244827671335 & 1005661010630 & 314913697573 & 1306500860355 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 288 factors with signed exponents of absolute value up to 1725940362787; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 31$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s - 482)\, y^3 + (20s - 44079)\, y^2 + (35s + 713966)\, y + (-1386s - 1930467)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 949*y^8 - 84278*y^7 + 6612677*y^6 - 151021673*y^5 + 2
786318041*y^4 - 41877012822*y^3 + 430627163226*y^2 - 3203861858505*y
+ 12564715259943
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + (-\frac{1443}{474207141581824})\, s, \qquad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3850075537 = 7 \cdot 11 \cdot 50000981$:

$$\begin{pmatrix} 3850075537 & 1434080588 & 1597579715 & 3833635359 & 257950722 & 3475155485 & 2580354442 & 2407147704 & 1276244565 & 3848950618 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 288 factors with signed exponents of absolute value up to 1106990938546; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 31$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s - 482)\, y^3 + (20s - 44079)\, y^2 + (35s + 713966)\, y + (-1386s - 1930467)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 - 949*y^8 - 84278*y^7 + 6612677*y^6 - 151021673*y^5 + 2
786318041*y^4 - 41877012822*y^3 + 430627163226*y^2 - 3203861858505*y
+ 12564715259943
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right) s, \qquad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9560434789675 = 5^2 \cdot 7^3 \cdot 11 \cdot 61^2 \cdot 27239$:

$$\begin{pmatrix}
639707915 & 364902244 & 223980176 & 540741395 & 95491380 & 576664442 & 362818452 & 4839848 & 212624880 & 161079383 \\
0 & 427 & 203 & 58 & 140 & 283 & 69 & 145 & 336 & 27 \\
0 & 0 & 35 & 10 & 1 & 24 & 24 & 25 & 10 & 22 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 291 factors with signed exponents of absolute value up to 1216145498765; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 31$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 641y^3 + (8s - 4)\, y^2 + 28924y + (448s - 224)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1282*y^8 + 468729*y^6 + 257277944*y^4 + 33804751120*y^2 + 923
108293632
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + (-\frac{7859}{2637314866215936})\, s, \qquad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3379297465600 = 2^8 \cdot 5^2 \cdot 7^3 \cdot 1539403$:

$$\begin{pmatrix}
3017229880 & 2265225060 & 722616516 & 2370831304 & 1481317588 & 629026663 & 1843216854 & 758218013 & 534913726 & 333008840 \\
0 & 70 & 54 & 12 & 55 & 18 & 67 & 58 & 59 & 42 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 4 & 0 & 2 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 164 factors with signed exponents of absolute value up to 5923241; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a+b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 641y^3 + (8s-4)\,y^2 + 28924y + (448s - 224)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1282*y^8 + 468729*y^6 + 257277944*y^4 + 33804751120*y^2 + 923
108293632
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right) s, \qquad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $63249306624 = 2^{10} \cdot 3^3 \cdot 7^2 \cdot 46687$:

$$\begin{pmatrix}
15686832 & 14612640 & 10998624 & 12818577 & 354075 & 9233536 & 2573589 & 6514586 & 9436875 & 13507881 \\
0 & 84 & 14 & 81 & 81 & 55 & 28 & 24 & 71 & 64 \\
0 & 0 & 4 & 3 & 1 & 1 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 3 & 1 & 1 & 2 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 159 factors with signed exponents of absolute value up to 2524112; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a+b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 641y^3 + (8s - 4)\,y^2 + 28924y + (448s - 224)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1282*y^8 + 468729*y^6 + 257277944*y^4 + 33804751120*y^2 + 923
108293632
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right) s, \qquad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $22226745936 = 2^4 \cdot 3 \cdot 47^2 \cdot 209623$:

$$\begin{pmatrix}
118227372 & 110213778 & 103294564 & 65726847 & 101011218 & 48267241 & 108578627 & 63915267 & 24632636 & 3108525 \\
0 & 94 & 0 & 93 & 84 & 0 & 74 & 67 & 39 & 92 \\
0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 158 factors with signed exponents of absolute value up to 11597937; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 1 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 169)\, y^3 + (-26s + 42585)\, y^2 + (-223s + 688385)\, y + (-2892s + 6182745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 341*y^8 + 84470*y^7 + 5834003*y^6 + 263122559*y^5 + 7
180070821*y^4 + 140622190062*y^3 + 1920411009606*y^2 + 1444123966095
9*y + 76675889982213
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + (-\frac{7859}{2637314866215936})\, s, \qquad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4632621291525 = 3 \cdot 5^2 \cdot 61768283887$:

$$\begin{pmatrix}
926524258305 & 726591467135 & 493991577802 & 319189311657 & 504916185110 & 743419938445 & 352533320822 & 749934338429 & 658490721791 & 737628397608 \\
0 & 5 & 2 & 0 & 4 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 351 factors with signed exponents of absolute value up to 913638474; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 169)\, y^3 + (-26s + 42585)\, y^2 + (-223s + 688385)\, y + (-2892s + 6182745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 341*y^8 + 84470*y^7 + 5834003*y^6 + 263122559*y^5 + 7
180070821*y^4 + 140622190062*y^3 + 1920411009606*y^2 + 1444123966095
9*y + 76675889982213
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right) s, \qquad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5610924323529 = 3 \cdot 7 \cdot 41 \cdot 271 \cdot 24047059$:

$$\begin{pmatrix}
5610924323529 & 2501272477059 & 3912479458208 & 3809475082778 & 4378846937718 & 1269781019844 & 360937245643 & 432259038917 & 2014765609147 & 4443095873087 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 394 factors with signed exponents of absolute value up to 2634332384; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 169)\, y^3 + (-26s + 42585)\, y^2 + (-223s + 688385)\, y + (-2892s + 6182745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):


```
y^10 - 4*y^9 + 341*y^8 + 84470*y^7 + 5834003*y^6 + 263122559*y^5 + 7
180070821*y^4 + 140622190062*y^3 + 1920411009606*y^2 + 1444123966095
9*y + 76675889982213
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + (-\frac{59454}{16272883861333507})\, s, \qquad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3224017064919 = 3 \cdot 7^2 \cdot 41 \cdot 9281 \cdot 57637$:

$$\begin{pmatrix} 460573866417 & 358499068724 & 139339085430 & 453081714576 & 406654364579 & 343315426363 & 179605807510 & 187420834880 & 350287547942 & 268376770279 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 7 & 1 & 0 & 6 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 359 factors with signed exponents of absolute value up to 9878942347; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 4)\, y^3 + (-8s - 37286)\, y^2 + (-163s + 361971)\, y + (1274s - 803530)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 7*y^8 - 74580*y^7 + 5323143*y^6 - 74934384*y^5 + 182980715*y^
4 - 3269841036*y^3 + 219290760724*y^2 - 2491338258864*y + 8109734610
432
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{373699}{329664358276992} + (-\frac{7859}{2637314866215936})\, s, \qquad J = \begin{pmatrix} 1884 & 1639 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1884$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18644750208 = 2^7 \cdot 3^3 \cdot 7 \cdot 797 \cdot 967$:

$$\begin{pmatrix} 129477432 & 95902242 & 108570576 & 97337992 & 119631291 & 629817 & 97699846 & 38209097 & 2736289 & 22706110 \\ 0 & 6 & 0 & 4 & 5 & 4 & 0 & 0 & 4 & 4 \\ 0 & 0 & 6 & 4 & 0 & 4 & 2 & 5 & 4 & 4 \\ 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 138 factors with signed exponents of absolute value up to 27395003; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s-4)\,y^3 + (-8s - 37286)\,y^2 + (-163s + 361971)\,y + (1274s - 803530)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 7*y^8 - 74580*y^7 + 5323143*y^6 - 74934384*y^5 + 182980715*y^
4 - 3269841036*y^3 + 219290760724*y^2 - 2491338258864*y + 8109734610
432
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{316613}{474207141581824} + \left(-\frac{1443}{474207141581824}\right) s, \qquad J = \begin{pmatrix} 1876 & 24 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1876$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4119743462400 = 2^{10} \cdot 3^6 \cdot 5^2 \cdot 31 \cdot 7121$:

$$\begin{pmatrix} 79470360 & 3739320 & 26734572 & 59723190 & 13670058 & 71995302 & 36562432 & 63543761 & 75496852 & 30112327 \\ 0 & 90 & 76 & 8 & 2 & 60 & 50 & 82 & 83 & 47 \\ 0 & 0 & 4 & 2 & 2 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 6 & 0 & 2 & 0 & 5 & 5 & 0 \\ 0 & 0 & 0 & 0 & 6 & 0 & 2 & 2 & 3 & 5 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 138 factors with signed exponents of absolute value up to 13045; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 4)\, y^3 + (-8s - 37286)\, y^2 + (-163s + 361971)\, y + (1274s - 803530)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 7*y^8 - 74580*y^7 + 5323143*y^6 - 74934384*y^5 + 182980715*y^
4 - 3269841036*y^3 + 219290760724*y^2 - 2491338258864*y + 8109734610
432
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{3927925}{16272883861333507} + \left(-\frac{59454}{16272883861333507}\right) s, \qquad J = \begin{pmatrix} 1747 & 1290 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1747$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2628553539584 = 2^{25} \cdot 7 \cdot 19^2 \cdot 31$:

$$\begin{pmatrix} 527744 & 48448 & 153216 & 252660 & 456196 & 353046 & 432639 & 292722 & 457558 & 294948 \\ 0 & 64 & 0 & 48 & 12 & 44 & 16 & 58 & 6 & 36 \\ 0 & 0 & 608 & 316 & 224 & 398 & 372 & 414 & 598 & 67 \\ 0 & 0 & 0 & 4 & 0 & 2 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 140 factors with signed exponents of absolute value up to 384994603; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.